Project Initialization and Planning Phase

Date	06 July 2026
Team ID	xxxxxx
Project Title	Smart Lender
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to assess and predict the creditworthiness of loan applicants using machine learning, boosting the efficiency of loan approval decisions for banks and financial institutions. It tackles the inefficiencies of manual credit evaluation, promising faster insights, consistent risk classification, and better decision-making tools. Key features include a machine learning-based classification model (Decision Tree, Random Forest, KNN, and XGBoost) and a real-time interactive Flask web interface for instant approval prediction.

Project Overview

Field	Details
Objective	The primary objective is to develop a predictive machine learning model that accurately classifies loan applicants as likely to repay or default, based on demographic, financial, and credit history indicators, and to select the best-performing model (XGBoost) for deployment.
Scope	The project comprehensively processes structured applicant data, trains and evaluates four classification algorithms, integrates the best model into a Flask web application, and provides a real-time interface for credit officers and financial analysts to evaluate loan applications, with deployment on IBM Cloud.

Problem Statement

Field	Details
Description	Addressing the inefficiency and inconsistency of manual creditworthiness assessment, which currently relies on subjective document review and rule-of-thumb judgement, especially for applicants with irregular self-employment income or limited credit history.
Impact	Solving these issues will result in faster loan approvals, reduced non-performing assets (NPAs), consistent and data-driven risk evaluation, and the ability to scale evaluation during high-volume application periods.

Proposed Solution

Field	Details
Approach	Employing supervised machine learning classification techniques (Decision Tree, Random Forest, KNN, and XGBoost) to analyze historical applicant records and predict loan repayment likelihood, then saving the best-performing model for real-time inference.
Key Features	• Implementation and comparison of four classification models, with XGBoost selected as the best performer (94.7% training accuracy, 81.1% testing accuracy). • Real-time decision-making via an interactive Flask web application. • Fast-track approval for low-risk applicants and automatic flagging of high-risk applicants for further scrutiny. • Support for batch evaluation of multiple applicants during high-volume periods.

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU specifications	Intel Core i3 or above
Memory	RAM specifications	Minimum 4 GB (8 GB recommended)
Storage	Disk space for data, models, and logs	Minimum 10 GB free space
Software		
Frameworks	Python web frameworks	Flask
Libraries	Additional libraries	scikit-learn, xgboost, pandas, numpy, matplotlib, seaborn
Development Environment	IDE	Jupyter Notebook, VS Code / PyCharm, Anaconda Navigator
Data	Source, size, format	Historical loan applicant records (CSV format) covering gender, marital status, education, employment, income, loan amount, loan term, credit history, and property area
Deployment	Cloud platform	IBM Cloud (Cloud Foundry / Watson Studio)